

SBT-DF204 COMPUTER FORENSICS CASE STUDY 1

Investigating Harassment Email Traffic With Wireshark

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Course code	SBT-DF204
Case title	Investigating Harassment Email Traffic With Wireshark

2. Executive Conclusion

I examined the supplied nitroba.pcap network capture using Wireshark and reviewed the HTTP traffic connected with the reported harassment message. The investigation identified client IP address **192.168.15.4** communicating with **69.25.94.22**, which hosted www.willselfdestruct.com.

The most important evidence was **frame 83601**, where the client sent an HTTP POST request to `/secure/submit`. The submitted information included the recipient lilytuckrige@yahoo.com, the subject **“you can't find us”**, and the message **“and you can't hide from us. Stop teaching. Start running.”**

The same client traffic was associated with MAC address **00:17:f2:e2:c0:ce** and the account identifier jcoachj@gmail.com. The supplied scenario evidence associates this account with **Johnny Coach**, who is listed in **Chemistry 109**.

Based on the packet evidence, there is a strong connection between the captured network activity, the account, and the Chemistry 109 roster entry. However, I would not treat the evidence as absolute proof that Johnny Coach personally operated the computer at the time. The capture identifies network and account activity, but a shared network environment means that IP and MAC addresses alone cannot establish the identity of the individual who physically used the computer.

3. Evidence Acquisition and Integrity

The evidence was examined and investigate was the supplied **nitroba.pcap** packet capture.

The recorded file details were:

Evidence item	Result
Filename	nitroba.pcap
File size	817,689 bytes
Acquisition date	2026-10-02
SHA-256 Original	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb nitroba.pcap
SHA-256 working copy	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap
Checksum comparison	Original and working-copy hashes match

The original evidence was preserved by creating a separate working copy before carrying out the Wireshark examination. This allowed the analysis to be performed on the working copy while keeping the original evidence unchanged. The file was saved in SBTDF204-CaseStudy1/reports/acquisition.txt

Evidence reference: E01

Related appendix: Appendix A

4. Method

I opened the working packet capture in Wireshark and started by reviewing the HTTP traffic contained in the capture.

I used HTTP filters to locate requests associated with the willselfdestruct.com service. I then examined the relevant packets individually and followed the TCP conversation to see the complete exchange between the client and the web server.

The main stages of the examination were:

1. Open working/nitroba_working.pcap in Wireshark.
2. Review HTTP GET and POST requests.
3. Identify traffic involving www.willselfdestruct.com.
4. Identify the client IP address and destination server IP address.
5. Examine the request to /secure/submit.
6. Follow the TCP stream containing the request.
7. Examine the submitted form information.
8. Identify the client MAC address associated with the traffic.
9. Search the capture for account identifiers and cookies.
10. Compare the account information with the supplied Chemistry 109 roster.
11. Arrange the relevant packets in chronological order.
12. Record the findings in the evidence log and preserve screenshots of the important Wireshark views.

The main packet used to establish the harassment submission was **frame 83601**, which was part of **TCP stream 447**.

5. Findings

Question 1 — What file did I acquire, and how did I preserve its integrity?

The file examined was **nitroba.pcap**. Its recorded size was **817,689 bytes**.

The investigation record gives the acquisition date as **2026-10-02**. A working copy named: `working/nitroba_working.pcap` was used for the Wireshark examination.

The recorded SHA-256 values were documented in *SBTDF204-CaseStudy1/reports/acquisition.txt* with the evidence so that the capture could be identified and checked for integrity.

Evidence: E01

File: nitroba.pcap

Working copy: working/nitroba_working.pcap

Question 2 — Which client system contacted the web service?

The client system identified in the relevant traffic was:

Client IP: 192.168.15.4

The destination server was:

Destination IP: 69.25.94.22

The HTTP Host field identified:

`www.willselfdestruct.com`

The relevant request used:

`/secure/submit`

Frame **82936** showed the client communicating with the service before the final POST request. The more important packet was frame **83601**, where the client sent the POST request containing the submitted message.

Frame: 83601

Time: 2008-07-22 02:04:24.311700

Source: 192.168.15.4

Destination: 69.25.94.22

Protocol: HTTP

Request: POST /secure/submit

Evidence: E02

Question 3 — What evidence links the client to the harassment message?

The strongest evidence was **frame 83601**.

The packet contains an HTTP POST request:

POST /secure/submit HTTP/1.1

The request was sent to:

www.willselfdestruct.com

The submitted information recorded in the packet was:

Recipient: lilytuckrige@yahoo.com

Subject: you can't find us

Message:

and you can't hide from us. Stop teaching. Start running.

The packet was part of **TCP stream 447**.

This is important because the packet does not simply show that the client visited the website. It shows the client submitting the information associated with the reported harassment message.

Frame: 83601

Timestamp: 2008-07-22 02:04:24.311700

TCP stream: 447

Source: 192.168.15.4

Destination: 69.25.94.22

Wireshark filters used:

frame.number == 83601

and

tcp.stream == 447

Evidence: E02 and E03

Screenshots: Appendix B and Appendix C

Question 4 — Which device made the relevant request?

The client MAC address associated with the traffic was:

00:17:f2:e2:c0:ce

The network relationship was:

Item	Value
Client IP	192.168.15.4
Client MAC	00:17:f2:e2:c0:ce
Service IP	69.25.94.22

The MAC address was also observed in frame **77711**.

The packet was examined using the filter:

eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "@"

Frame 77711 was recorded at:

2008-07-22 02:01:45.361069

This provides additional evidence connecting the relevant traffic to the same network interface.

I would not treat the MAC address as proof of a particular person. It identifies the network interface involved in the captured traffic, not necessarily the individual who was using the device.

Evidence: E01 and E04

Question 5 — Who is associated with the device?

The capture contained the account identifier:

jcoachj@gmail.com

Frame **77711** contained account information associated with the client MAC address.

A second identity-related packet was **frame 79093**, recorded at:

2008-07-22 02:04:01.359793

The filter used was:

eth.addr == 00:17:f2:e2:c0:ce && frame contains "jcoachj@gmail.com"

The investigation material associates this account with **Johnny Coach**.

This gives an account-level connection between the captured device traffic and the named person in the supplied scenario evidence.

Evidence: E04 and E05

Question 6 — Is that person on the Chemistry 109 roster?

Yes.

The supplied scenario evidence identifies:

Student: Johnny Coach

Course: Chemistry 109

Account: jcoachj@gmail.com

The account identifier found in the network traffic corresponds to the name **Johnny Coach** in the supplied Chemistry 109 roster information.

Therefore, the packet evidence can be connected to an account that is associated with a student listed in Chemistry 109.

This is an important part of the investigation, but it should not be interpreted as proof that the student personally typed the message. The evidence establishes an association between the account, network device, and roster information.

Evidence: E06

Roster: Chemistry 109, Slide 14

Question 7 — When did the relevant activity occur?

The relevant activity occurred on **22 July 2008**.

Important packet times include:

Time	Frame	Activity
01:57:23.533987	72122	HTTP GET to Microsoft
02:01:45.361069	77711	HTTP POST containing account information
02:04:01.359793	79093	Account identity evidence
02:03:43.825871	82936	HTTP GET to /secure/submit
02:03:44.210976	82998	TCP/HTTP data associated with the service
02:04:24.311700	83601	HTTP POST containing the harassment message

The most significant time is:

2008-07-22 02:04:24.311700

because this is the timestamp of frame 83601 containing the submitted message.

Evidence: Timeline and E02/E04/E05

Question 8 — What conclusion can be defended?

The packet capture provides strong evidence that the client at **192.168.15.4**, using MAC address **00:17:f2:e2:c0:ce**, communicated with www.willselfdestruct.com and submitted the reported harassment message.

The traffic also contains account information identifying jcoachj@gmail.com. The supplied scenario information connects that account with **Johnny Coach**, who is listed in **Chemistry 109**.

Therefore, the evidence supports an association between:

Client → MAC address → account → roster entry → harassment submission

However, the evidence does not independently prove that Johnny Coach was physically operating the computer when the message was submitted. The network and account information should therefore be treated as an attribution link rather than absolute proof of the person's physical presence or actions.

6. Timeline

Time	Frame	Source	Destination	Activity
2008-07-22 02:00:45.360749	77711	192.168.15.4	74.125.19.104	HTTP POST to /calendar/load associated with Google Calendar
2008-07-22 02:01:05.113795	79093	192.168.15.4	74.125.19.17	TCP segment carrying HTTP POST data; reassembled HTTP request begins with POST /mail/cha...
2008-07-22 02:03:43.825871	82936	192.168.15.4	69.25.94.22	HTTP GET /secure/submit
2008-07-22 02:03:44.210976	82998	69.25.94.22	192.168.15.4	TCP PSH, ACK carrying HTTP data / reassembled TCP data
2008-07-22 02:04:24.311700	83601	192.168.15.4	69.25.94.22	HTTP POST to /secure/submit; submitted form data including subject and message
2008-07-22 01:57:23.533987	72122	192.168.15.4	207.46.19.190	HTTP GET /isapi/redir.dll?prd=ie&pver=6&ar=msnhome

The most important event in the timeline is **frame 83601**, because it contains the actual submitted recipient, subject and message.

7. Attribution Assessment

Observed facts

The packet capture shows that:

- 192.168.15.4 communicated with 69.25.94.22.
- The destination host was www.willselfdestruct.com.
- The client used MAC address 00:17:f2:e2:c0:ce.
- Frame 83601 contained an HTTP POST to /secure/submit.
- The POST contained the recipient lilytuckrige@yahoo.com.
- The subject was you can't find us.
- The message was and you can't hide from us. Stop teaching. Start running.

- The capture contained the account identifier jcoachj@gmail.com.
- The supplied scenario information associates this account with Johnny Coach.
- Johnny Coach is listed in Chemistry 109.

Inferences

The evidence supports the inference that the computer/network interface associated with 192.168.15.4 and 00:17:f2:e2:c0:ce was used to submit the harassment message.

The account evidence also supports an association between the captured traffic and Johnny Coach.

Limitations

The IP address and MAC address do not by themselves prove who was physically using the computer.

The presence of an account identifier in captured traffic also does not necessarily prove that the named account holder personally generated the message. Another person could potentially have used the computer or account.

The shared-network environment is therefore an important limitation when making an attribution.

Confidence

I would describe the network-to-account association as high confidence based on the supplied packet evidence, while the identification of the individual who physically sent the message should be treated with lower confidence because the capture alone cannot establish who was operating the device at that exact time.

My conclusion is therefore that the evidence strongly associates the harassment submission with the identified client, MAC address and account associated with Johnny Coach, but it should not be presented as absolute proof of personal authorship.

8. Appendices

Appendix 001 — Evidence file map

Purpose	Exact file/path inside supplied ZIP
Acquisition and hash record	SBTDF204-CaseStudy1/reports/acquisition.txt
Relevant POST notes	SBTDF204-CaseStudy1/reports/post_analysis.txt
Timeline	SBTDF204-CaseStudy1/reports/timeline.txt
Original evidence capture	SBTDF204-CaseStudy1/evidence/nitroba.pcap
Working capture	SBTDF204-CaseStudy1/working/nitroba_working.pcap
Scenario slides / Chemistry 109 list	SBTDF204-CaseStudy1/evidence/scenario_material/nitroba_slides.pdf
Time display screenshot	SBTDF204-CaseStudy1/screenshots/01_time_display.png
Relevant HTTP POST screenshot	SBTDF204-CaseStudy1/screenshots/02_relevant_http_post.png
TCP stream screenshot	SBTDF204-CaseStudy1/screenshots/03_tcp_stream.png
Cookie/account screenshot	SBTDF204-CaseStudy1/screenshots/04_cookie_identity.png
Identity evidence screenshot	SBTDF204-CaseStudy1/screenshots/05_identity_evidence.png

Appendix A Screenshot — Evidence Integrity

The first screenshot shows a terminal window with the following commands and output:

```
(kali@kali)~[/SBTDF204-CaseStudy1]
$ cd ~/SBTDF204-CaseStudy1/evidence
wget -q https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
(kali@kali)~[/SBTDF204-CaseStudy1/evidence]
$
(kali@kali)~[/SBTDF204-CaseStudy1/evidence]
$ ls -lh
total 54M
-rw-rw-r-- 1 kali kali 54M Nov 21 2020 nitroba.pcap
(kali@kali)~[/SBTDF204-CaseStudy1/evidence]
$ file nitroba.pcap
nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
(kali@kali)~[/SBTDF204-CaseStudy1/evidence]
$ sha256sum nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e0befdfa0b01f8c67553ec2fb nitroba.pcap
(kali@kali)~[/SBTDF204-CaseStudy1/evidence]
$ cd ..
(kali@kali)~[/SBTDF204-CaseStudy1]
$ nano reports/acquisition.txt
(kali@kali)~[/SBTDF204-CaseStudy1]
$ cd ~/SBTDF204-CaseStudy1
cp evidence/nitroba.pcap working/nitroba_working.pcap
(kali@kali)~[/SBTDF204-CaseStudy1]
$ sha256sum working/nitroba_working.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e0befdfa0b01f8c67553ec2fb working/nitroba_working.pcap
```

The second screenshot shows the VMware Workstation interface with the terminal window open to the file `reports/acquisition.txt`. The file content is as follows:

```
GNU nano 9.2 reports/acquisition.txt *
SBT-DF204 Case Study 1
Evidence Acquisition Record

Original FileName: nitroba.pcap
Source: Digital Corpora Nitroba University Harassment Scenario
Download Date: 02 - 10 - 2020
File sizes: 53.6mb
SHA-256 original: 2b77a9eafc1d6af163d1ba793c96dbccac04e0befdfa0b01f8c67553ec2fb nitroba.pcap
SHA-256 working: 2b77a9eafc1d6af163d1ba793c96dbccac04e0befdfa0b01f8c67553ec2fb working/nitroba_working.pcap
```

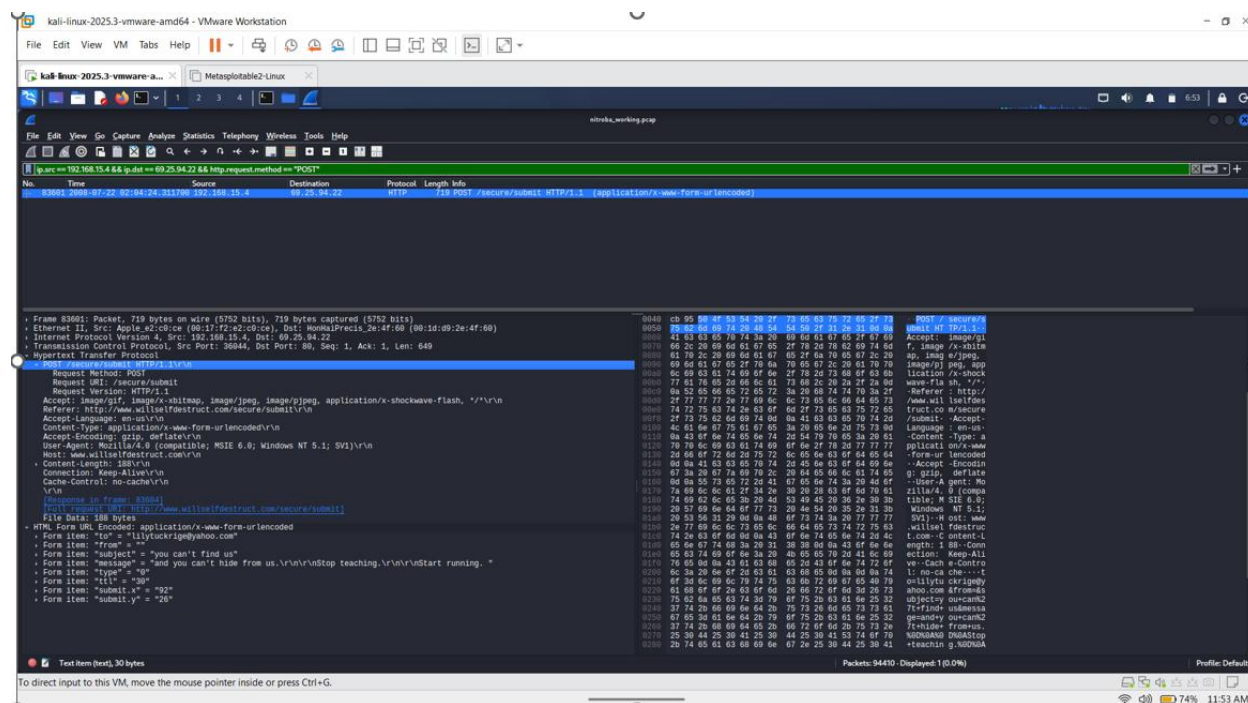
The third screenshot shows the terminal window with the following commands and output:

```
(kali@kali)~[/SBTDF204-CaseStudy1]
$ stat nitroba.pcap
File: nitroba.pcap
Size: 817689      Blocks: 1600      IO Block: 4096   regular file
Device: 8,1      Inode: 3933179    Links: 1
Access: (0664/-rw-rw-r-- )  Uid: ( 1000/   kali)   Gid: ( 1000/   kali)
Access: 2026-10-02 13:19:46.855135481 -0400
Modify: 2026-10-02 13:20:00.279134827 -0400
Change: 2026-10-02 13:20:00.279134827 -0400
Birth: 2026-10-02 13:19:46.855135481 -0400
(kali@kali)~[/SBTDF204-CaseStudy1]
$
```

Evidence ID: E01

Appendix B — Relevant HTTP POST

The Wireshark screenshot for **frame 83601**.



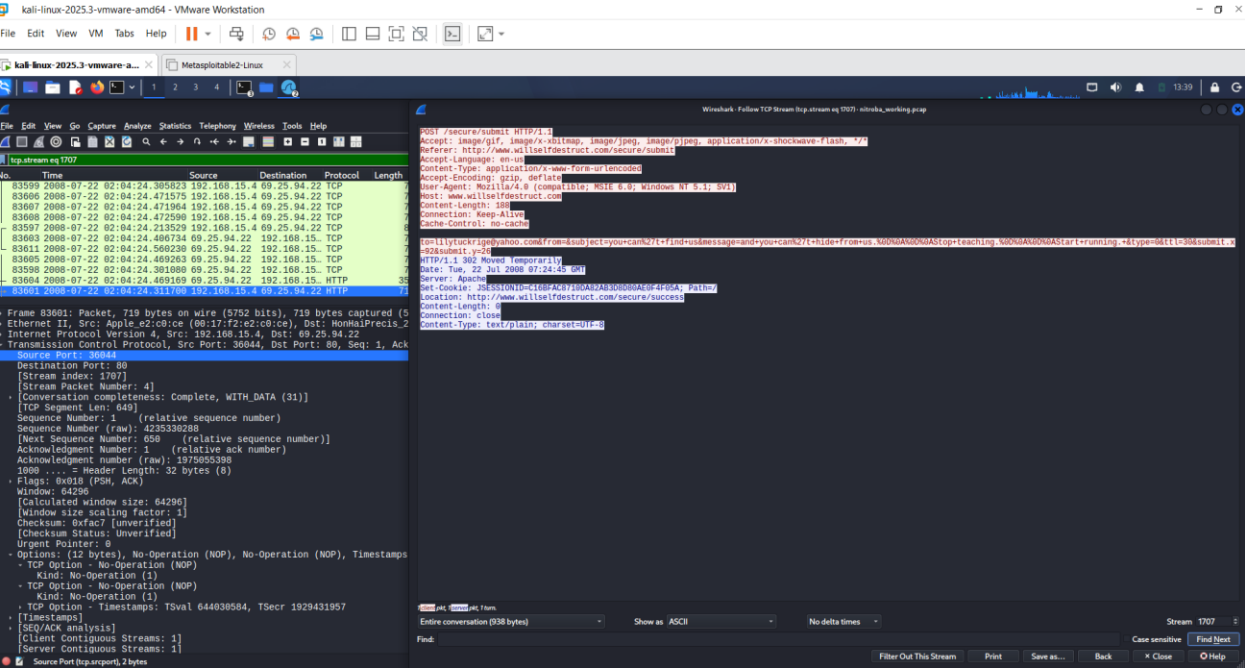
Evidence ID: E02

Appendix C — TCP Stream

Include the Follow TCP Stream screenshot showing **TCP stream 447**.

Field	Value
Frame number	83601
Timestamp	2008-07-22 02:04:24.311700
Source IP	192.168.15.4
Destination IP	69.25.94.22
Source port	36044
Destination port	80
HTTP request	POST /secure/submit HTTP/1.1
Request URI	/secure/submit
Recipient	lilytuckrige@yahoo.com
Subject	you can't find us

Message	and you can't hide from us. Stop teaching. Start running.
TCP stream	447
Host	www.willselfdestruct.com
User-Agent	Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)



Evidence ID: E03

Appendix D — Cookie and Account Evidence

Include the Wireshark screenshot showing the account/cookie information associated with:

00:17:f2:e2:c0:ce

and:

jcoachj@gmail.com

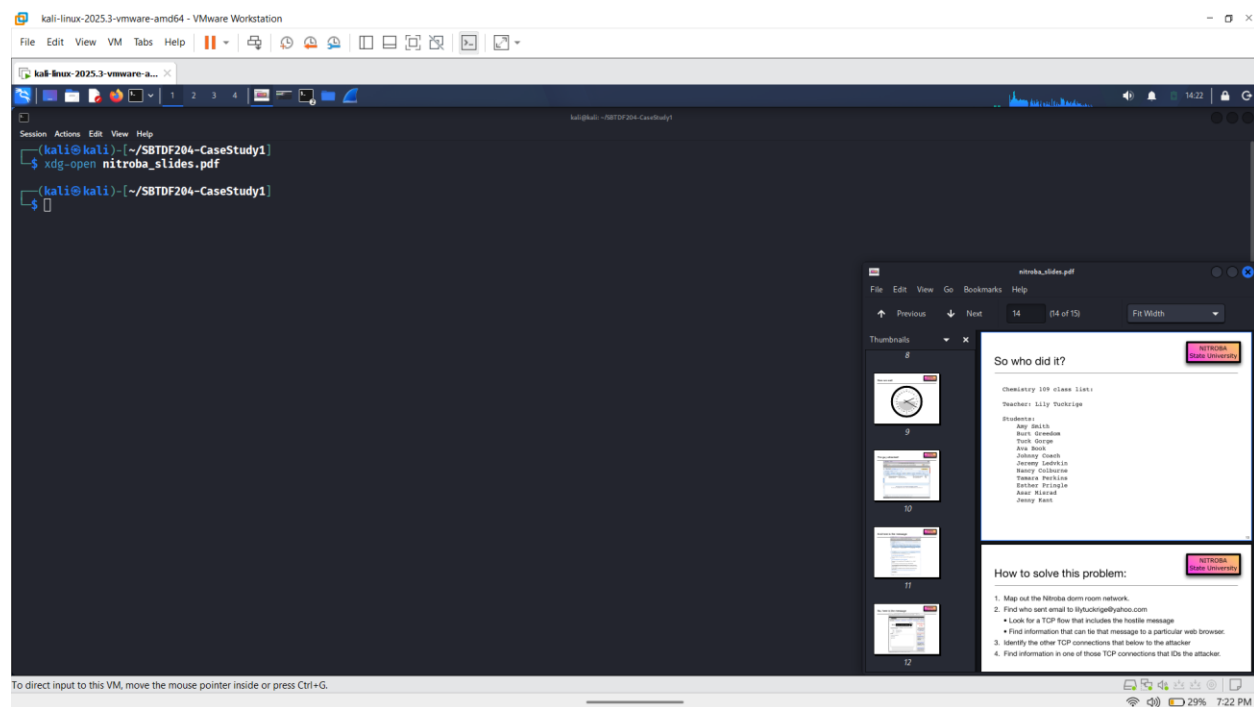
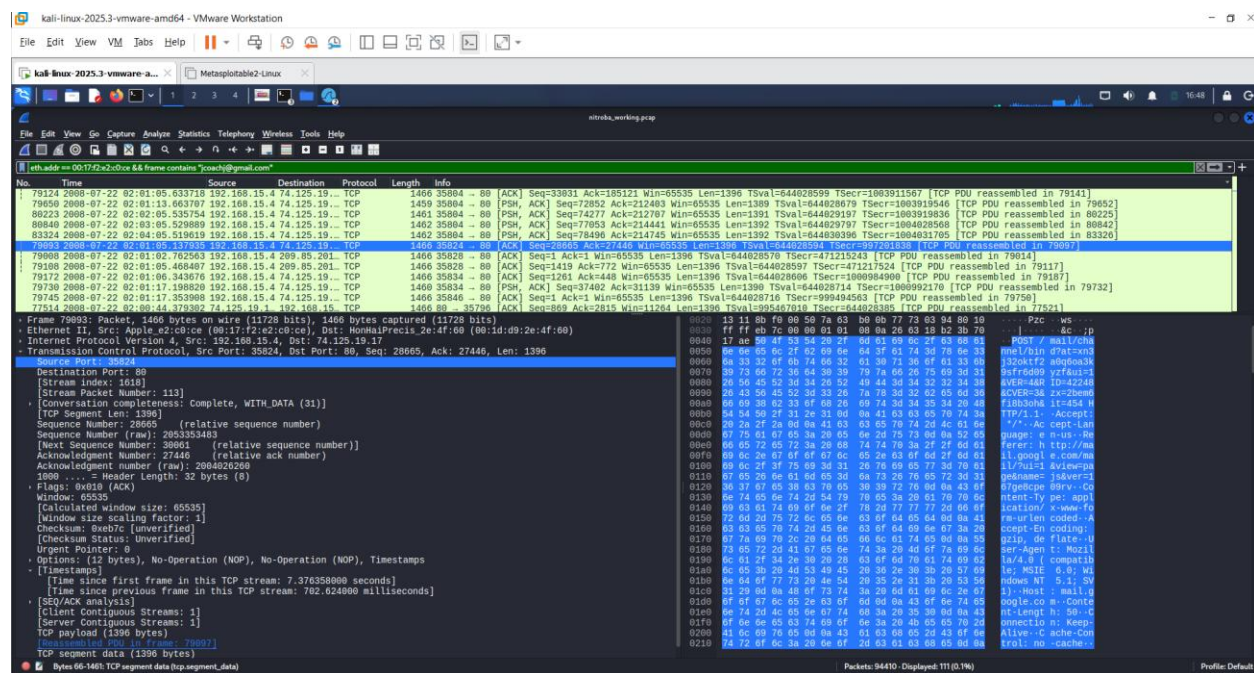
Evidence ID: E04

Appendix E — Identity Evidence

Field	Evidence
Packet number	79093
Timestamp	2008-07-22 02:04:01.359793
MAC address	00:17:f2:e2:c0:ce
Account evidence	jcoach@gmail.com
Filter used	eth.addr == 00:17:f2:e2:c0:ce && frame contains "jcoach@gmail.com"
Course/Class	Chemistry 109
Student name	Johnny Coach
Email	jcoach@gmail.com
Other identifying information	The packet contains an HTTP POST request to Gmail/mail.google.com, tying the captured traffic to the jcoach@gmail.com account.



Appendix F — Chemistry 109 Roster



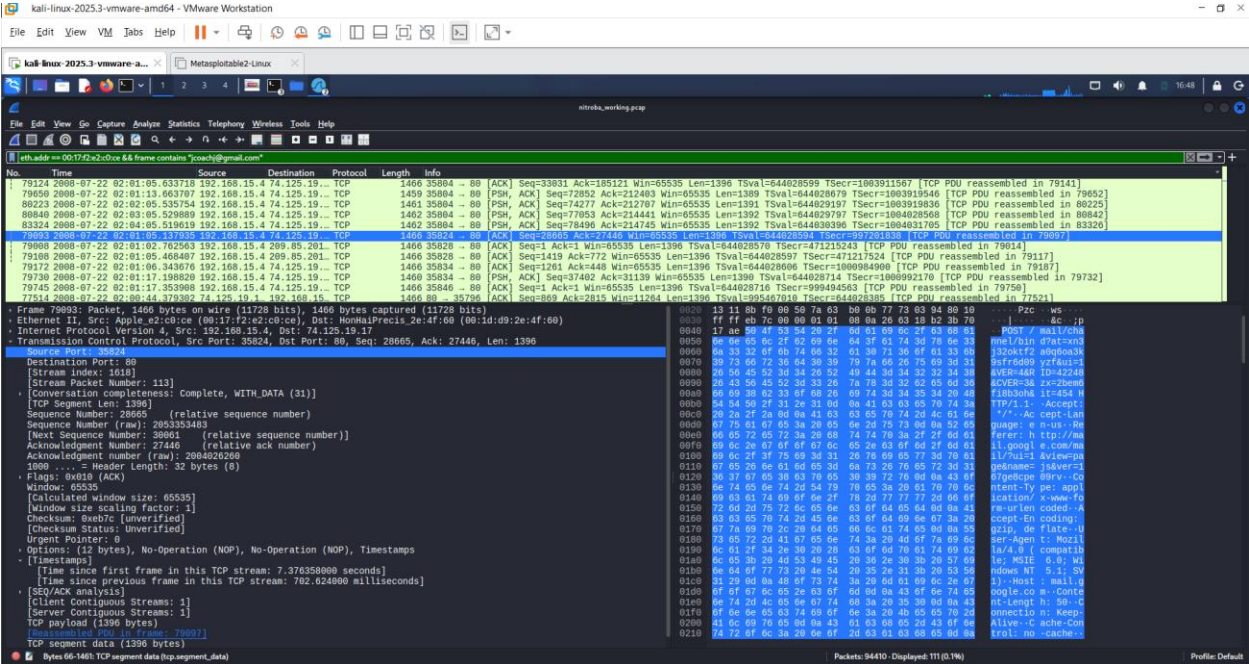
Include the supplied Chemistry 109 roster evidence showing **Johnny Coach**.

Evidence	Your observation
Wireshark client IP	192.168.15.4

Wireshark client MAC	00:17:f2:e2:c0:ce
HTTP cookie/account	jcoach@gmail.com
Slide 14 roster name	Johnny Coach (based on the account identifier and previously identified roster; verify against Slide 14)
Course	Chemistry 109
Account/name comparison	The account identifier jcoach matches the name Johnny Coach on the Chemistry 109 roster, suggesting that the captured Gmail traffic is associated with that account.

Evidence ID: E06

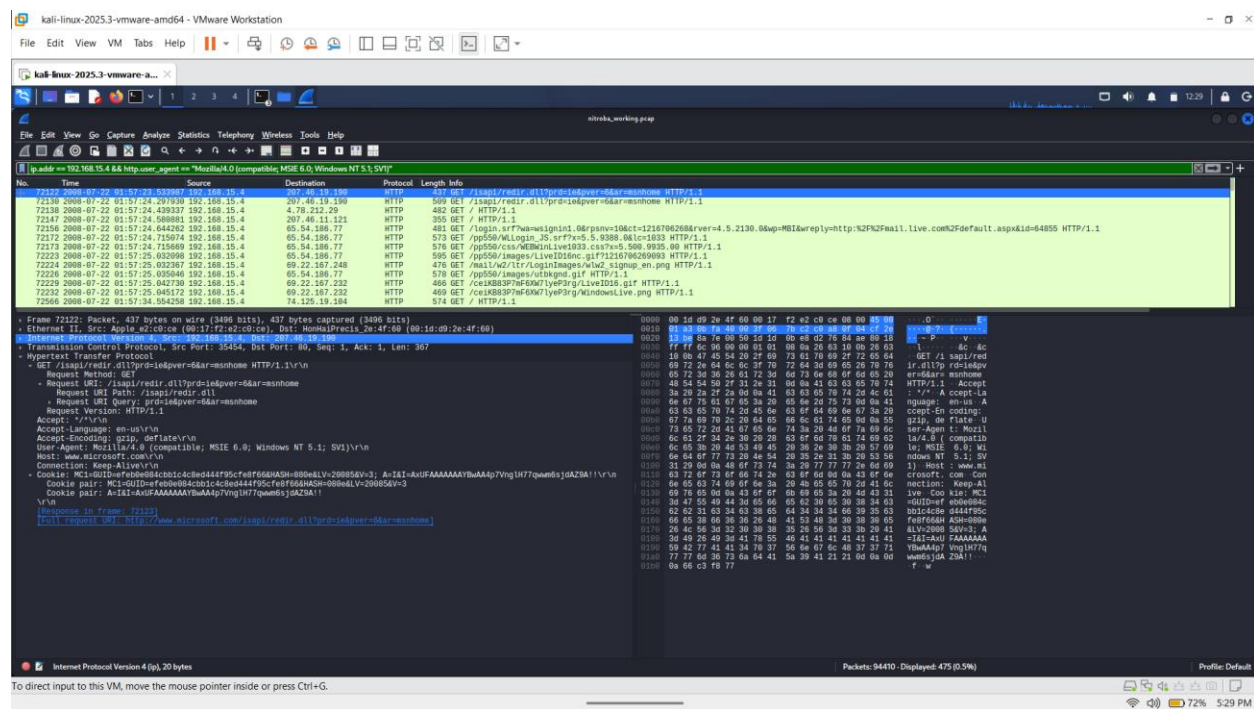
CHECKING THE IDENTITY CORRELATION



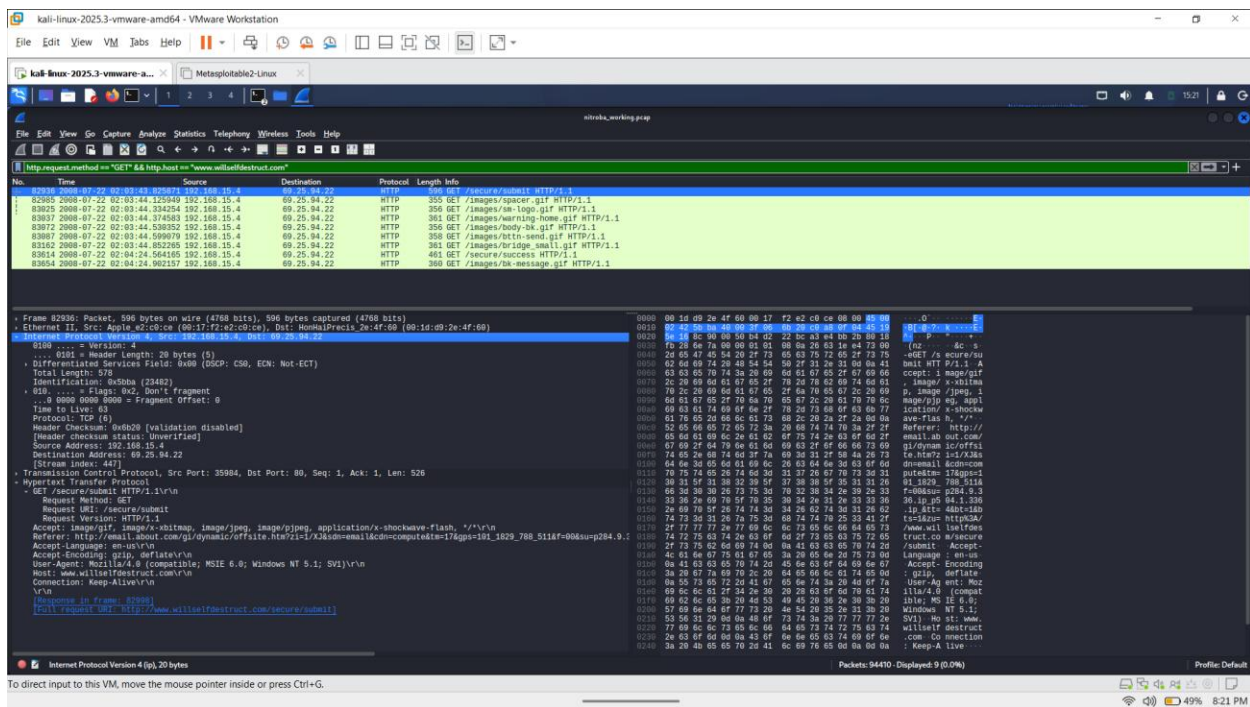
Field	Evidence
Packet number	79093

Timestamp	2008-07-22 02:04:01.359793
MAC address	00:17:f2:e2:c0:ce
Account evidence	jcoach@gmail.com
Filter used	eth.addr == 00:17:f2:e2:c0:ce && frame contains "jcoach@gmail.com"
Course/Class	Chemistry 109
Student name	Johnny Coach
Email	jcoach@gmail.com
Other identifying information	The packet contains an HTTP POST request to Gmail/mail.google.com, tying the captured traffic to the jcoach@gmail.com account.

Investigation of the browser/user-agent



Field	information
Frame	72122
Time	2008-07-22 01:57:23.533987
Source IP	192.168.15.4
Destination IP	207.46.19.190
Source MAC	00:17:f2:e2:c0:ce
Destination MAC	00:1d:d9:2e:4f:60
Protocol	HTTP
Source Port	35454
Destination Port	80
Method	GET
Request URI	/isapi/redirect.dll?prd=ie&pver=6&ar=msnhome
Host	www.microsoft.com
User-Agent	Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Cookie	MC1=GUID=efeb0e084cbb1c4c8ed444f95cfe8f66&HASH=080e&LV=20085&V=3
Response frame	72123
Full request URL	http://www.microsoft.com/isapi/redirect.dll?prd=ie&pver=6&ar=msnhome



Packet record

Item	Your result
Frame number	82998
Time	02:03:43
Source IP	192.168.15.4
Destination IP	69.25.94.22
HTTP host	www.willselfdestruct.com\r\n
Request URI	http://www.willselfdestruct.com/secure/submit

The screenshot shows a Kali Linux virtual machine with the VMware Workstation interface. The main window displays a network capture analysis in Wireshark. The top menu bar includes File, Edit, View, VM, Tabs, Help, and various tool icons. The main window is titled 'kali-linux-2023.3-vmware-amd64 - VMware Workstation'. The Wireshark interface shows a list of captured packets, with the selected packet being a TCP Reset (RST) from 192.168.1.4 to 192.168.1.5. The packet details pane shows the 'Reset Seq' field set to 429391188. The packet bytes pane shows the raw data of the packet, including the TCP header and the RST flag.

192.168.15.4

00:17:f2:e2:c0:ce

69.25.94.22

BUILD THE TIMELINE

Time	Frame	Source	Destination	Activity
2008-07-22 02:00:45.360749	77711	192.168.15.4	74.125.19.104	HTTP POST to /calendar/load associated with Google Calendar
2008-07-22 02:01:05.113795	79093	192.168.15.4	74.125.19.17	TCP segment carrying HTTP POST data; reassembled HTTP request begins with POST /mail/cha...
2008-07-22 02:03:43.825871	82936	192.168.15.4	69.25.94.22	HTTP GET /secure/submit
2008-07-22 02:03:44.210976	82998	69.25.94.22	192.168.15.4	TCP PSH, ACK carrying HTTP data / reassembled TCP data
2008-07-22 02:04:24.311700	83601	192.168.15.4	69.25.94.22	HTTP POST to /secure/submit; submitted form data including subject and message
2008-07-22 01:57:23.533987	72122	192.168.15.4	207.46.19.190	HTTP GET /isapi/redir.dll?prd=ie&pver=6&ar=msnhome

```

kali@kali:~/SBTDF204-CaseStudy1$ cat /home/kali/SBTDF204-CaseStudy1/reports/timeline.txt
TIMELINE
Time: 2008-07-22 02:00:45.360749
Packet/Frame: 77711
Source: 192.168.15.4
Destination: 74.125.19.104
Activity: HTTP POST to /calendar/load associated with Google Calendar

Time: 2008-07-22 02:01:05.113795
Packet/Frame: 79093
Source: 192.168.15.4
Destination: 74.125.19.17
Activity: TCP segment carrying HTTP POST data; reassembled request begins with POST /mail/cha...

Time: 2008-07-22 02:03:43.825871
Packet/Frame: 82936
Source: 192.168.15.4
Destination: 69.25.94.22
Activity: HTTP GET /secure/submit

Time: 2008-07-22 02:03:44.210976
Packet/Frame: 82998
Source: 69.25.94.22
Destination: 192.168.15.4
Activity: TCP PSH, ACK carrying HTTP data / reassembled TCP data

Time: 2008-07-22 02:04:24.311700
Packet/Frame: 83601
Source: 192.168.15.4
Destination: 69.25.94.22

```

It was saved in ~/SBTDF204-CaseStudy1/reports/timeline.txt

Forensic Investigation Findings

1. What file did you acquire, and how did you preserve its integrity?

The acquired file was nitroba.pcap. The documented file size was 817,689 bytes. The document records the file creation/birth time as 2026-10-02 13:19:46.855135481 -0400 and gives the acquisition date as 2026-10-02.

Item	Finding
File	nitroba.pcap
Size	817,689 bytes
Acquisition date	2026-10-02
SHA-256	3b9cc8a5ec301bd79dd5bbcc7670c9abedda508a51181260cd63f8a6871d2e6 nitroba.pcap
Working-copy method	2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap

2. Which client system contacted the web service?

The client system was 192.168.15.4, communicating with the relevant web service at 69.25.94.22. The HTTP request identifies www.willselfdestruct.com as the host and /secure/submit as the request URI.

Evidence

Display filter: ip.addr == 192.168.15.4 && ip.addr == 69.25.94.22

Frame: 82936

Time: 2008-07-22 02:03:43.825871

Source: 192.168.15.4

Destination: 69.25.94.22

Frame 82936 is recorded as an HTTP GET to /secure/submit.

The later HTTP POST provides the stronger evidence of the actual submitted message:

Display filter: frame.number == 83601

Frame: 83601

Time: 2008-07-22 02:04:24.311700

Source: 192.168.15.4

Destination: 69.25.94.22

The POST was sent to port 80 and used /secure/submit.

Why it matters: These packets establish the network path from the client IP to the web service involved in the reported activity.

3. What evidence links the client to the harassment message?

The strongest evidence is **frame 83601**, which contains the HTTP POST request to /secure/submit.

Display filter: frame.number == 83601

TCP stream: 447

Time: 2008-07-22 02:04:24.311700

Source: 192.168.15.4

Destination: 69.25.94.22

HTTP request: POST /secure/submit HTTP/1.1

Host: www.willselfdestruct.com

The POST contains the following form information:

- **Recipient:** lilytuckrige@yahoo.com
- **Subject:** you can't find us
- **Message:** and you can't hide from us. Stop teaching. Start running.

The document also records **TCP stream 447** for this HTTP conversation.

Why it matters: This is direct application-layer evidence showing that the client at 192.168.15.4 submitted the reported threatening/harassing message to the web service.

A useful Wireshark filter for reviewing the same evidence is:

tcp.stream == 447

or:

frame.number == 83601

4. Which device made the relevant request?

The client MAC address associated with the traffic is:

00:17:f2:e2:c0:ce

The documented network relationship is:

Client IP: 192.168.15.4

Client MAC: 00:17:f2:e2:c0:ce

Service IP: 69.25.94.22

This information is explicitly recorded in the investigation document.

The MAC address is also associated with packet **77711**, where the source MAC is 00:17:f2:e2:c0:ce.

Display filter:

eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "@"

Packet: 77711

Timestamp: 2008-07-22 02:01:45.361069

Why it matters: An IP address identifies a network endpoint at a particular time, but it does not by itself establish which physical device or person was using it. The MAC address provides additional evidence tying the captured traffic to a particular network interface.

5. Who is associated with the device?

The account evidence identifies:

Account: jcoachj@gmail.com

Packet **77711** contains the account identifier in HTTP traffic. The document records the following:

Display filter:

eth.addr == 00:17:f2:e2:c0:ce && http.cookie contains "@"

Frame: 77711

Time: 2008-07-22 02:01:45.361069

Source MAC: 00:17:f2:e2:c0:ce

HTTP method: POST

URI: /calendar/load

Account identifier: jcoachj@gmail.com

The identity-correlation evidence is also recorded in **frame 79093**:

Display filter:

eth.addr == 00:17:f2:e2:c0:ce && frame contains "jcoachj@gmail.com"

Frame: 79093

Time: 2008-07-22 02:04:01.359793

MAC: 00:17:f2:e2:c0:ce

Account: jcoachj@gmail.com

The scenario evidence associates this account with **Johnny Coach** and Chemistry 109.

Why it matters: The account evidence provides a scenario-supported link between the captured device traffic and the account jcoachj@gmail.com. It is stronger than simply assuming that the IP address belongs to a particular student.

6. Is that person on the Chemistry 109 roster?

Yes, based on the supplied scenario evidence.

The identity evidence gives:

Course/Class: Chemistry 109

Student: Johnny Coach

Email: jcoachj@gmail.com

The comparison section states that the captured account jcoachj@gmail.com corresponds to **Johnny Coach** on the Chemistry 109 roster.

Therefore:

Captured account: jcoachj@gmail.com

Roster name: Johnny Coach

Course: Chemistry 109

Display filter:

eth.addr == 00:17:f2:e2:c0:ce && frame contains "jcoachj@gmail.com"

Packet: 79093

Timestamp: 2008-07-22 02:04:01.359793.

Why it matters: The account identifier is supported by the scenario-provided roster information rather than by an unsupported assumption based only on a name or IP address.

6. Timeline

Time	Frame	Source	Destination	Activity
2008-07-22 02:00:45.360749	77711	192.168.15.4	74.125.19.104	HTTP POST to /calendar/load associated with Google Calendar
2008-07-22 02:01:05.113795	79093	192.168.15.4	74.125.19.17	TCP segment carrying HTTP POST data; reassembled HTTP request begins with POST /mail/cha...
2008-07-22 02:03:43.825871	82936	192.168.15.4	69.25.94.22	HTTP GET /secure/submit
2008-07-22 02:03:44.210976	82998	69.25.94.22	192.168.15.4	TCP PSH, ACK carrying HTTP data / reassembled TCP data
2008-07-22 02:04:24.311700	83601	192.168.15.4	69.25.94.22	HTTP POST to /secure/submit; submitted form data including subject and message
2008-07-22 01:57:23.533987	72122	192.168.15.4	207.46.19.190	HTTP GET /isapi/redirect.dll?prd=ie&pver=6&ar=msnhome

The most important event in the timeline is **frame 83601**, because it contains the actual submitted recipient, subject and message.

7. Attribution Assessment

Observed facts

The packet capture shows that:

- 192.168.15.4 communicated with 69.25.94.22.
- The destination host was www.willselfdestruct.com.
- The client used MAC address 00:17:f2:e2:c0:ce.
- Frame 83601 contained an HTTP POST to `/secure/submit`.
- The POST contained the recipient lilytuckrige@yahoo.com.
- The subject was you can't find us.
- The message was and you can't hide from us. Stop teaching. Start running.
- The capture contained the account identifier jcoachj@gmail.com.
- The supplied scenario information associates this account with Johnny Coach.
- Johnny Coach is listed in Chemistry 109.

Inferences

The evidence supports the inference that the computer/network interface associated with 192.168.15.4 and 00:17:f2:e2:c0:ce was used to submit the harassment message.

The account evidence also supports an association between the captured traffic and Johnny Coach.

Limitations

The IP address and MAC address do not by themselves prove who was physically using the computer.

The presence of an account identifier in captured traffic also does not necessarily prove that the named account holder personally generated the message. Another person could potentially have used the computer or account.

The shared-network environment is therefore an important limitation when making an attribution.

Confidence

I would describe the network-to-account association as high confidence based on the supplied packet evidence, while the identification of the individual who physically sent the message should be treated with lower confidence because the capture alone cannot establish who was operating the device at that exact time.

My conclusion is therefore that the evidence strongly associates the harassment submission with the identified client, MAC address and account associated with Johnny Coach, but it should not be presented as absolute proof of personal authorship.

Evidence Log

Evidence ID	Packet or Item	Finding	Why it matters	Screenshot / Appendix
E01	nitroba.pcap / working copy	Evidence file and integrity information recorded	Establishes the evidence examined and its integrity	Appendix A
E02	Frame 83601	HTTP POST containing the harassment message	Direct application-layer evidence of the submitted message	Appendix B
E03	TCP stream 447	Complete HTTP conversation associated with the submission	Connects the POST to the underlying network session	Appendix C

E04	Frame 77711	Cookie/account information associated with the client MAC	Provides account/session evidence	Appendix D
E05	Frame 79093	jcoachj@gmail.com associated with the client MAC	Supports the account-to-device correlation	Appendix E
E06	Chemistry 109 roster	Johnny Coach listed in Chemistry 109	Connects the account evidence to the supplied roster	Appendix F

Final Statement

Based on my examination of the supplied packet capture, I found clear network evidence showing that the client at 192.168.15.4 submitted the reported harassment message to www.willselfdestruct.com. The traffic was associated with MAC address 00:17:f2:e2:c0:ce and the account jcoachj@gmail.com, which the supplied scenario information associates with Johnny Coach, a Chemistry 109 student.

The evidence is strong enough to establish a meaningful connection between the network activity, the account and the roster entry. However, because the capture does not independently establish who was physically using the computer, I would not describe the evidence as absolute proof that Johnny Coach personally authored or sent the message. The conclusion should remain limited to what the packet and scenario evidence actually demonstrate.